ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

FACT CHECK · C9 · HEAT TRANSFER

Conduction, convection or radiation?

★☆☆
Grade 5

Learning goals

- Identify the type of heat transfer.
- Classify conductors and insulators.

1 Which kind of heat transfer is it? [6]

Example	Conduction, convection or radiation?
A metal pan handle gets hot.	conduction
Warm air rises from a heater.	convection
The Sun warms your face.	radiation
Your feet feel cold on a tiled floor.	conduction
A hot-air balloon rises.	convection
You feel warm in front of a fire.	radiation

2 Sort these materials. [6]

SORT THESE

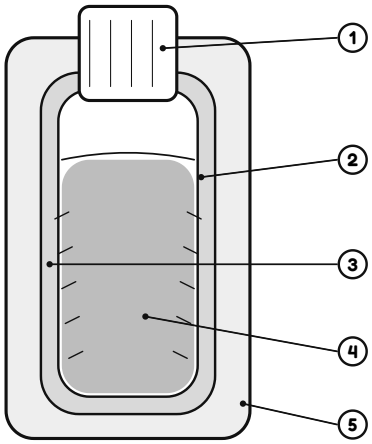
steel **copper** **wool** **trapped air** **aluminium** **wood**

Good conductor	Good insulator
copper	wool
steel	wood
aluminium	trapped air

3 **Label the vacuum flask.** [5]

LABELS

stopper **shiny silver walls** **hot drink** **vacuum (empty space)** **outer case**



1 stopper

2 shiny silver walls

3 vacuum (empty space)

4 hot drink

5 outer case

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

THINK IT THROUGH · C9 · HEAT TRANSFER

Explaining heat in daily life

★★☆
Grade 5

Learning goals

- Explain with the particle model.
- Apply ideas to design.

1 **Explain, using particles, why a metal spoon in hot dal gets hot all the way up.** [3]

The particles at the hot end vibrate faster and bump into their neighbours, passing energy along the metal to the handle (conduction). Metal is a good conductor.

2 **Why are heaters put near the floor but air conditioners high on the wall?** [3]

Warm air rises, so a low heater sets up a convection current that warms the whole room; cool air sinks, so a high air conditioner cools the room as the cold air falls.

3 **Why do people in hot countries often wear white or light-coloured clothes?** [2]

Light colours reflect radiation from the Sun, so they absorb less heat.

4 **A cook's pan has a copper base and a plastic handle. Explain both choices.** [2]

Copper conducts heat quickly to the food; plastic is an insulator so the handle stays cool.

ANSWER KEY

INVESTIGATE · C9 · HEAT TRANSFER

Which colour heats up fastest in the sun?

★★☆ Grade 5

Learning goals

- Plan a fair test.
- Plot and interpret a line graph.

Class 5B wrapped three identical cans of water in black paper, white paper and shiny foil. They put them in the sun and measured the water temperature every 10 minutes.

1 **Name the variables.** [3]

a) Changed:

the colour/covering of the can

b) Measured:

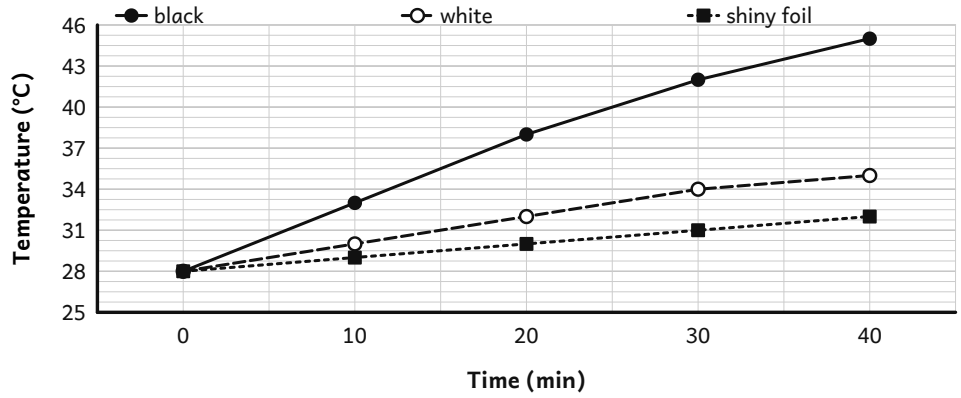
the water temperature

c) Kept the same:

same cans, same amount of water, same starting temperature, same place in the sun, same times

2 **Plot the three lines on one graph (use a key).** [4]

Time (min)	0	10	20	30	40
black (°C)	28	33	38	42	45
white (°C)	28	30	32	34	35
shiny foil (°C)	28	29	30	31	32



3 **Which heated fastest? Explain why using the word radiation.** [2]

The black can: dark, dull surfaces absorb radiation from the Sun best. Shiny foil reflects most of it.

4 **Use your results to choose a colour for a water tank on a roof that should stay cool.** [2]

White or shiny, because it reflects radiation and absorbs least heat.

Data: example results for practice.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REAL-WORLD CASE STUDY · C9 · HEAT TRANSFER

Keeping cool in a hotter Dhaka

★★★
Grade 5

Learning goals

- Apply heat transfer to a real problem.
- Compare solutions using data.

Science | Design and technology | Health | Maths

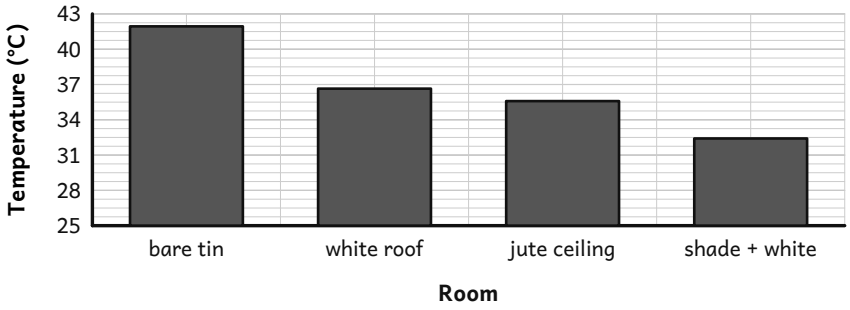
Many homes in Dhaka have thin **tin (corrugated iron) roofs**. On a sunny day the dark metal absorbs radiation and conducts heat straight into the room below. Inside, it can be much hotter than outside, which is dangerous for babies and older people during a heatwave.

Some families have tried cheap ideas: painting the roof **white**, adding a **ceiling of jute or bamboo** with an air gap, and growing **creepers or trees** for shade. A community group measured the temperature inside four rooms at 2 pm on the same hot day.

Example data based on cool-roof trials in hot cities; numbers are illustrative.

1 **Draw a bar chart of the results.** [3]

Room	bare tin roof	white-painted roof	jute ceiling with air gap	tree shade + white roof
Temperature at 2 pm (°C)	41	36	35	32



2 **Explain why the white roof was cooler, using radiation.** [2]

White reflects more of the Sun's radiation, so the roof absorbs less heat.

3 **Explain why the jute ceiling with an air gap helped, using conduction.** [2]

Jute and trapped air are insulators, so less heat is conducted from the hot roof into the room.

4 **Recommend one solution for a poor family. Give a scientific and a money reason.** [3]

e.g. white paint: cheap, quick, reflects radiation; or plant a tree: free shade, also cleans the air.

ANSWER KEY

InquiryMind
LEARN · PLAY · WONDER

REVIEW AND CHALLENGE · C9 · HEAT TRANSFER

Show what you know

★★★
Grade 5

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 **Find the words.**

S	O	L	K	A	Y	W	C	H	I	T
R	A	D	I	A	T	I	O	N	I	T
I	I	J	U	T	V	L	N	S	H	H
N	A	H	Y	K	H	K	V	N	G	J
S	M	H	E	A	T	U	E	I	H	W
U	V	A	C	U	U	M	C	A	K	K
L	C	O	N	D	U	C	T	I	O	N
A	N	H	J	W	P	U	I	I	R	U
T	M	E	T	A	L	P	O	J	A	H
O	F	C	B	L	G	M	N	Y	W	V
R	J	V	P	H	H	C	L	S	J	D

conduction **convection** **heat** **insulator** **metal** **radiation** **vacuum**

2 **Heat travels through empty space by...** [1]

◯ conduction ◯ convection

● **radiation**

3 **Which is the best insulator?** [1]

◯ copper ◯ steel

● **wool**

4 **Challenge. Design a lunch box that keeps rice hot for 4 hours. Label how it reduces conduction, convection and radiation.** [4]

My hot lunch box

e.g. double walls with an air gap (conduction), a tight lid (convection), shiny inner lining (radiation), wool cover.